Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.18	20.10	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.01	-0.4%
power (W), mean	627	630	+0.5%
power (W), peak	656	674	+2.7%
energy (Wh)	211	211	+0.1%
energy, parked workers still on at idle power (Wh)	211	211	+0.1%
CPU used (cores), mean	0.843	0.922	+9.4%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.035	0.038	+9.4%
energy per core-hour (Wh)	744	684	-8.2%
node-hours per core-hour	7.12	6.51	-8.6%
pending pods, pod-minutes	0.000	0.250	new
pending pods, peak	0.000	1.00	new
HPA replicas, mean	8.85	9.04	+2.2%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	1.05	0.767	-27.0%
requests timed	1,070	1,150	+7.5%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	123	47.78	-61.1%
response time (ms), median	112	42.40	-62.1%
response time (ms), 95th percentile	252	71.70	-71.5%
response time (ms), 99th percentile	333	93.57	-71.9%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.8426	7.4440	-5.1%	[-2.5462, +1.7122]	EVEN, no significant difference
kwh_per_core_hour	0.8162	0.7768	-4.8%	[-0.2419, +0.1650]	EVEN, no significant difference
utilisation	0.0348	0.0380	+9.3%	[-0.0072, +0.0145]	EVEN, no significant difference
pending_pod_minutes_per_hour	0.0000	0.7200	n/a	[+0.0000, +2.1600]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	3.0360	2.2439	-26.1%	[-5.2918, +3.7198]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.392 0.42 0.444 0.466 0.485 0.501 0.516 0.531 0.542 0.551 0.557 0.563 0.567 0.572 0.577 0.581 0.584 0.585 0.587
- nodes decided per minute: 6 6 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	614	4.75	5.00	0.000	0.000	0.528	0.474

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	615	620	5.00	5.75	0.000	0.000	0.517	0.629
2	6.00	6.00	615	615	5.00	6.00	0.000	0.000	0.508	0.511
3	6.00	6.00	622	620	5.25	6.00	0.000	0.250	0.705	0.658
4	6.00	6.00	632	636	7.50	8.50	0.000	0.000	0.968	1.06
5	6.00	6.00	628	634	10.00	10.00	0.000	0.000	0.890	1.05
6	6.00	6.00	634	636	10.00	10.00	0.000	0.000	1.00	1.07
7	6.00	6.00	646	660	10.00	10.00	0.000	0.000	1.31	1.69
8	6.00	6.00	648	657	10.00	10.00	0.000	0.000	1.40	1.62
9	6.00	6.00	646	654	10.00	10.00	0.000	0.000	1.34	1.56
10	6.00	6.00	639	650	10.00	10.00	0.000	0.000	1.15	1.46
11	6.00	6.00	616	618	10.00	10.00	0.000	0.000	0.574	0.579
12	6.00	6.00	617	615	10.00	10.00	0.000	0.000	0.563	0.505
13	6.00	6.00	621	622	10.00	10.00	0.000	0.000	0.688	0.686
14	6.00	6.00	632	636	10.00	10.00	0.000	0.000	0.936	1.09
15	6.00	6.00	633	635	10.00	10.00	0.000	0.000	0.991	1.05
16	6.00	6.00	633	635	10.00	10.00	0.000	0.000	1.05	1.02
17	6.00	6.00	625	625	10.00	10.00	0.000	0.000	0.746	0.794
18	6.00	6.00	616	618	10.00	10.00	0.000	0.000	0.549	0.566
19	6.00	6.00	616	616	10.00	10.00	0.000	0.000	0.553	0.541

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.